MILITARY SPECIFICATION

HYPODERMIC INJECTION APPARATUS, JET, FOOT OPERATED

This specification is approved for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1.1 Scope This specification covers one type of hydraulically operated, hypodermic, jet injection apparatus for rapidly injecting, at high pressure and velocity, through a jet orifice, metered quantities of immunizing agents in such a manner that the immunizing agent (vaccine) makes its own opening into the skin and penetrates to a suitable depth for immunization.

2 APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein.

SPECIFICATIONS

GG-N-196	Needles, Hypodermic, for Luer Syringes.
QQ-A-367	Aluminum Alloy Forgings
QQ-S-763	Steel Bars, Shapes, and Forgings-Corrosion Resisting.
QQ-S-766	Steel Plates, Sheet, and Strip-Corrosion Resisting.
QQ-W-423	Wire, Steel, Corrosion-Resisting.
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-P-116	Preservation-Packaging, Methods of.
MIL-W-530	Webbing, Textile, Cotton, General Purpose, Natural or In Colors.
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-H-9890	Hardware, Individual Food Carrying Equipment, and Hardware, Miscellaneous
MIL-F-10884	Fasteners, Snap.
MIL-P-15024	Plate, Tags and Bands For Identification of Equipment.
MIL-G-23827	Grease, Aircraft and Instrument, Gear and Actuator Screw.

FSC 6515

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STANDARDS

Air Force - Navy Aeronautical

AN936	Washer, Tooth Lock
AN6227	Packing, O-Ring Hydraulic
AN6270	Hose Assembly-Detachable End Fitting, Low Pressure

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MS16686	Medical Services Symbol
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure
MS28782	Retainer, Packing, Back-up Teflon

DRAWING

Defense Personnel Support Center

25000	Hypodermic Injection Apparatus, Jet Foot Powered and Bill of Materials

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS
ASTM B140 Leaded Nickel Brass (Leaded Nickel Silver) and Leaded Nickel Bronze (Leaded Nickel Silver) Sand Castings

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

AMERICAN IRON AND STEEL INSTITUTE
Steel Products Manual, Stainless and Heat-Resisting Steels

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, New York 10017).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
ANSI B46.1 Surface Texture.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

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3 REQUIREMENTS

3.1 Material. Except as otherwise authorized by the contracting officer, material shall be as specified in the bill of materials accompanying Defense Personnel Support Center (DPSC) Drawing 25000 and as specified herein. Material shall be free of all defects and imperfections that may affect the serviceability of the finished product.

3.2 Proprietary components. When trade-named items or other proprietary components are specified, products of other manufacturers may be substituted provided that they are equal to, and interchangeable with, the item specified, and written approval of the contracting officer has been obtained.

3.3 Construction. The hypodermic injection apparatus shall be constructed strictly in accordance with Drawing 25000, and the requirements described herein. The hypodermic injection apparatus shall consist of two main subassemblies (1) an injection "gun" and (2) a hydraulic power source. These two components shall be interconnected by means of hydraulic hoses and shall be housed within a plastic carrying case. Complete jet injection apparatus shall pass the performance test specified in 4.3.6.

3.3.1 Injection "gun". The injection "gun" shall consist of the component parts assembled as shown in sheet 2 of the drawing.

3.3.1.1 Stock. The aluminum-alloy gun stock (part 11) shall be accurately fabricated to meet the dimensional tolerances shown on the detail drawing. Alignment of passageways, valves, and ports shall be such that all components function smoothly, without hydraulic leakage to cock the main spring and to fire the injection. All stepped holes shall be concentric. Surfaces on which seals ride shall be equal to, or superior to, the finish specified in the drawing (see 3.3.5.1). Observation windows, calibrated from 0 to 1.0 cubic centimeters (cc), in 0.1 cc subdivisions, shall be provided on each side of the stock. The exterior surface of the stock shall be given a satin finish (approximately 280 grit) prior to anodizing and waxed after anodizing. Particular care shall be taken to insure that rear surface of the grip of the stock is well rounded for operator comfort. All surfaces of the stock shall be given a black anodized finish as specified in 3.3.5.2. While supporting the stock in the anodizing bath, particular care shall be taken not to grasp the stock at a finished surface. The top opening in the stock shall be sealed by means of a pin plug (part 80) after the stock has been anodized. The calibration marks and identification marking as specified in 3.3.4.1 shall be filled with a white pigment after anodizing.

3.3.1.2 Valves and plug. The stock of the "gun" shall be fitted with two valves (a firing valve and a check valve) and a plug. Valve components shall be constructed as shown in the detail drawing. When the foot pump is actuated, the hydraulic fluid shall be forced (via the check valve) to acutate the hydraulic piston (part 24A) and compress the main spring (part 85). The firing valve

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shall permit the operator to release the energy stored in the main spring by opening the pressure line to the hydraulic reservoir. The check valve shall trap the fluid holding the cocked spring in place after the pump is no longer actuated. The ball seats on the two valves shall be properly formed and free of defects so that there is no fluid leakage past the balls when the hydraulic pressure output of the pump is 1300 pounds per square inch (psi). The plug (part 15) shall seal the by-pass which is used to unload the electric-driven pump when the injector is a component part of NSN 6515-00-656-1021. There shall be no creeping of the hydraulic piston (part 24A) as seen through the observation window when the unit is given a leakage test as specified in 4.3.1.

3.3.1.3 Firing spring. The firing spring (part 85) shall supply the energy required to administer the injection. The firing spring shall be made of a chrome-vanadium steel formed to have the following properties:

Type - Compression die spring
Works in hole diameter - 1 inch
Works over rod diameter - 1/2 inch
Free length - 2-1/2 inches
Total deflection - 30% free length
Maximum load - 600 pounds
Load per 1/8 inch deflection - 85 pounds
End - closed and ground

Spring shall be the Raymond Manufacturing Company Die Spring, Catalog No. 1"x2-1/2" HD, or equal. The firing spring shall work over the spring guide portion of the hydraulic piston (part 24A), which shall be constructed from corrosion-resisting steel as shown in the drawing.

3.3.1.4 Hydraulic actuating device. The hydraulic piston (part 24A), seal (part 92), seal (part 91), and cylinder portion of the stock shall function to provide a hydraulic actuating device to compress the main spring. Fabricated parts shall be constructed as shown in the drawing while purchased parts shall be as described in the bill of materials accompanying the drawing. The outside diameter of the quad ring seals, parts 90, 91 and 92, shall be identified, by such means as color coding, to simplify their proper installation. If color coding is used, the coding material shall be non-toxic and non-permanent. Seals shall function so as to prevent leakage of hydraulic fluid from the gun stock observation windows or through the rear of the dosage adjusting screw while the injector is in use (see 4.3.6). A thin film of oil due to normal lubrication of cylinder walls as the hydraulic piston moves is normal and is not cause for rejection.

3.3.1.5 Dosage adjustment subassembly. Dosage adjustment from 0.1 to 1.0 cc shall be accomplished by means of the dosage adjustment subassembly comprising parts 26, 27, 28, 38, 81, 82 and 99. The threads of the adjusting stem (part 27) shall be coated with Molykote, type G. The thrust bearing (part 81), as well as the main spring (part 85), shall be coated with grease conforming to MIL-G-23827 prior to securing of this subassembly to the injection gun stock.

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3.3.1.6 Vaccine pump section. The vaccine pump section shall be a self-priming, single action pump (part 23), consisting of the following component parts: inlet valve housing (part 29), inlet valve (part 30), outlet valve retainer (part 31), outlet valve (part 32), cylinder (part 33), coupling nut (part 34), inlet tube (part 35), inlet valve mounting pin (part 36), key pin (part 79), valve balls (part 88), outlet valve spring (part 40), and seals (parts 90, 93, 94, 96, 97)

Fabricated parts shall be constructed as shown in the drawing, while purchase parts shall be as described in the bill of materials accompanying the drawings. The quad ring seal, part 90, shall be fabricated of material suitable for repeated pressure steam sterilizations at 250° F. Cylinder (part 33), shall be concentric with the bore of the stock to insure accurate alignment of the injection piston (part 23) with the hydraulic piston (part 24A). Surfaces on which seals ride shall be equal to, or superior to, the finish specified in the drawing. Keying of the vaccine pump section to the gun stock shall be accomplished by a pin (part 79) in such a manner that the inlet valve assembly in a vertical alignment with the handle of the gun stock. The ball seats of the pump inlet and outlet valves shall be properly formed and free of defects so that positive seals are obtained during normal operation of the injector. Inlet and outlet valves shall not leak when unit is tested as specified in 4.3.2.2. Tube (part 35) shall be securely brazed at both ends in an airtight, watertight manner. There shall be no trace of air bubbles leaking past the brazed joint, when the unit is tested as specified in 4.3.2.1.

3.3.1.6.1 Jet nozzle. The jet nozzle shall consist of corrosion-resisting steel blank (part 41) and a sapphire jewel orifice (part 42). The sapphire jewel orifice shall conform to the dimensions shown in the drawing when the jewel contour is checked by contour projection at 100 magnifications as described in 4.3.4. The mounting blank shall be fabricated in accordance with the detail drawing. The front face of the blank shall have a minimum thickness of 0.072 inch. Inside face of the blank shall have a surface equal to, or superior to, the finish specified in the drawing. The mounting hole for the sapphire jewel shall be concentric with the inside diameter of the jewel mount. The mounting of the sapphire jewel shall be by means of a friction setting, in which the bore of the housing is a minimum of 0.0002 inch smaller than the outside diameter of the jewel, followed by spinning of the blank material over the chamber of the jewel edge. The orifice of the jewel shall be perfectly clean, free of lapping or polishing materials. The orifice opening of the jewel shall be regular and circular in contour and shall be free of chips or other imperfections. The face of the jewel shall be flush with the face of the blank within the dimensions specified in the drawing. The face of the mount shall be machined to provide a non-slip medium diamond knurl radial pattern.

3.3.1.6.1.1 Protective cap. Each jet nozzle shall be provided with a protective cap fabricated from black, anodized finish, aluminum alloy (see 3.3.5.2). Cap shall be designed to properly protect jet nozzle. The length of the cap shall be 5/8 ± 1/64 inch.

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3.3.1.6.1.2 Nozzle cleaning adapters. Each apparatus shall be furnished with one nozzle cleaning adapter to clean clogged subcutaneous nozzle specified in 3.3.1.6.1 and one nozzle cleaning adapter to clean clogged intradermal nozzles (not furnished with this apparatus) NSN 6515-00-913-7908 The nozzle cleaning adapters shall be constructed to properly align with the orifice of the appropriate nozzle. The cleaning adapters assemblies shall be as shown on sheet 12 of DPSC Drawing 25000. Appropriate detailed instructions for use of the cleaning adapters shall be provided with the apparatus.

3.3.1.6.2 Feed needle assembly. The feed needle assembly shall consist of the following component parts: vaccine feed tube (part 44), vent tube (part 45), vaccine tube plug (part 46), Luer hubs (parts 83, 84), filter screen (part 84A), and Nylon adapter (part 73). Assembly of concentric tubes and of tubes to hubs shall be by means of sound silver-brazed joints, employing the brazing material specified in the drawing. The brazed joints shall be airtight when feed needle is tested in accordance with the procedure given in 4.3.3. Particular care shall be taken during the brazing operation not to distort the Luer hub (part 84) or to destroy the internal finish of this hub. Construction shall be such that an airtight joint can be made between the Luer hub (part 84) and the matching taper on the inlet valve (part 30). The filter screen (part 84A) shall be a 100 mesh monel wire screen approximately 3/16 inch diameter, suitably domed to fit securely into the vaccine inlet hub of each feed needle assembly.

3.3.1.7 Bottle holder assembly. The bottle holder assembly shall consist of the following component parts, stud (part 39), positioner (part 50), clip (part 51), jaws (part 52), jaw reinforcements (part 55), latch (part 53), and spring (part 54). Parts shall be fabricated in accordance with the details shown in the drawing. Sheet metal parts may be fabricated either of the alloys as shown in the bill of materials or of type AISI 430 stainless steel. The bottle holder shall be coated with a vinyl plastisol coating material as specified in 3.3.5.3. Coating shall be applied prior to attachment of parts 39, 53 and 54. The inside surface of part 50 shall be uncoated. The jaws shall securely grasp all sizes of vaccine bottles from 10 to 60 cc capacity. The finger-operated latch (part 53) shall operate smoothly, without binding, on the stud (part 39). The coated surface of part 51 which contacts the gun stock (part 11) shall be smoothly finished so that sliding of the bottle holder assembly back and forth on the gun stock shall not mar the black anodized finish of the stock. All exposed stainless steel parts shall be ground to a No. 4 uniform satin finish. The bottle holder assembly shall be designed to adequately support small and large vaccine bottles by Velcro hook tape (part 116) to the jaws (part 52). The support shall be completed by assembling Velcro loop tape (part 117) to the Velcro hook tape to form a loop. This assembly shown on sheet 11 of DPSC Drawing.

3.3.1.8 Multipurpose tool. Each injection unit shall be furnished with a corrosion-resistant steel multi-purpose tool (part 48) constructed as shown in the drawing. The pin end of the tool shall be capable of removing and inserting the valve (part 12) into the gun stock without burring, or otherwise damaging, this valve. The tool shall be capable of being slipped into the dosage adjustment knob (part 28) to afford additional leverage to the operator in making dosage adjustments.

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3.3.2 Hydraulic system. The hydraulic system shall consist of a hydraulic power unit, hydraulic fittings, and hoses.

3.3.2.1 Hydraulic power unit. The hydraulic power unit shall consist of a single compact unit comprising a foot operated single-action piston type pump, equipped with inlet and discharge check valves, an integral reservoir and an internal relief valve. The power unit shall be the Oil-Dyne, Inc. model FP-JIA. The power unit shall be of style and design shown in sheet 1 of the drawing and shall be provided with a mounting hole pattern to mate with the cast pump mount, part 17. The power unit shall be dimensioned so that when mounted it will fit within the carrying case, in the case area designated in the drawing, without interferance with other components. Nominal dimensions of the power unit shall be as follows:

Overall length (pedal depressed)	10 inches
Overall height (pump body)	3 inches
Overall height (pedal extended)	5 inches
Body diameter	3-1/4 inches

The maximum weight of the complete power unit, less pump mount, hoses and hydraulic fluid shall be 2 pounds, 9 ounces. The power unit components and the assembly shall pass the acceptance test specified in 4.3.5.

3.3.2.1.1 Hydraulic pump. The foot powered hydraulic pump shall be of the high pressure piston type capable of continuous duty at 3000 psi and up to 5000 psi peak. At all pressures up to 3000 psi, the pump shall be capable of delivering a minimum of 0.2 cubic inches of hydraulic fluid per stroke. When operating at a output pressure of 1300 psi, and connected to the jet injection apparatus by means of specified tubing and fittings, one complete stroke of the pump shall be capable of cocking the injector to a dose not less than 0.5 cc, as viewed at the injector observation window. The pedal of the pump is to be of such shape as to permit comfortable actuation by the operator's foot. The stroke of the pedal shall have a nominal value of 3-9/16 inches. The pump shall have an internal spring return permitting pedal to return to the extended position without mechanical shock. The maximum load to start or to complete a stroke at various relief valve settings shall be as tabulated below:

Relief Valve Setting, psi	Maximum Load Pounds
500	38
1000	63
1500	95
2000	131

The pedal shall not distort or otherwise be damaged when it is depressed and subjected to a load to 250 pounds. The hydraulic pump outlets shall come off the left side of the pump body (as viewed from pedal) and ports shall be suitably machined to accept the "O" ring seal of the male connectors, part 14, without

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leakage. All exposed aluminum alloy components shall be of black anodized finish (see 3.3.5.2) and all screws, shafts, shaft retaining rings, fittings, and other exposed parts shall be fabricated of corrosion-resisting steel. When mounted in the carrying case, the pump unit shall be capable of being cleaned by hosing with water without causing corrosion to components or otherwise having its function impaired.

3.3.2.1.2 Hydraulic reservoir. The hydraulic reservoir shall be an integral part of the hydraulic pump. The fluid capacity of the hydraulic reservoir shall be such that the power unit shall be capable of delivering six cubic inches of hydraulic fluid from the discharge port while no fluid is being returned to the reservoir through the return port. The reservoir shall be provided with a breather to compensate for expansion and contraction of hydraulic fluid due to temperature changes and for the piston rod displacement of the hydraulic piston, part 24A, within the injection apparatus. A diaphragm shall seal the breather from the reservoir thereby yielding a fully sealed type system permitting the power unit to be held or shipped in any position without fluid leaking from the breather. An adequate size filling port fitted with an "O" ring seal plug, shall be provided, so located that the reservoir is filled to level when pump unit is placed on its side and fluid is added into filler port until fluid appears at the neck of the opening.

3.3.2.1.3 Relief valve. The relief valve shall be mounted internally and shall only be capable of adjustment if the pump cover and diaphragm are removed. No special tools shall be required to make an adjustment. The relief valve shall be fitted with a corrosion-resisting steel spring capable of being adjusted so that the relief valve operates in the 500 to 2300 psi range. The relief valve shall be factory-set at 1300 ± 50 psi. The relief valve shall actuate at ± 8 percent of the set pressure.

3.3.2.1.4 Hydraulic fluid. The hydraulic fluid shall be Oil Dyne O-D 18 Hydraulic Fluid, or equal, and shall be capable of being used in ambients ranging from -30° F. to 120° F. The storage temperature range shall be -60° F. to 150° F. Operating altitude condition shall be from -500 feet to 10,000 feet referred to sea level.

3.3.2.1.5 Hydraulic connections and hoses. Connection of the injection gun to the hydraulic power unit shall be by means of the male connectors, part 14. No compound shall be used on the threads. Before being connected to the jet injection apparatus hydraulic hose assemblies shall be flushed with hydraulic fluid by connecting to a hydraulic pump with an output capacity at least equal to the capacity of the hydraulic pump specified herein. Flushing shall continue until no foreign matter is present in flushed fluid Hoses shall be sealed after assembly and resealed after flushing to minimize possible entry of foreign matter into the hydraulic system.

3.3.3 Carrying case. The carrying case shall be molded from Royalite ABS #26 Morrocan Grain, Royalite, U.S. Rubber Company, or equal, as shown on sheets 1, 3, 4, and 5 of the drawing. A draft angle of 5° in molding is permitted. The case bottom and top shall be fitted with mating aluminum alloy extruded valance. The case top shall be hinged to the bottom by a continuous hinge riveted in place through the case valance as shown on the drawing. The case shall be provided with a stay hinge, a plastic handle and two surface locks, all located as shown on the drawing. The case top shall have a recessed section for securing of an identifying nameplate (see 3.3.4.2). On the inside of the top there shall be

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provided two gun clips (parts 7 and 8), a securing strap assembly (part 9) and a neoprene sponge cushion (part 69) which shall securely hold the jet injector in place when complete unit is subjected to the vibration test (see 3.11.8). Integral with the case bottom there shall be molded a cavity which shall function as the body of a repair parts compartment. The cover of the compartment (part 3) shall be hinged in place by means of a corrosion-resisting steel continuous hinge (part 4) held in place by means of blind rivets. The cover shall be secured in place by means of a strap assembly (part 10). On the inside of the case bottom, in designated areas, there shall be provided the following components:

a. Clip (part 62) for multipurpose tool (part 48). Clip shall be riveted in place and kept from rotating by means of lock washer.

b. Aluminum-alloy (black anodized) spare injection mounts, 3 required (part 47), secured by rivets to inside of repair compartment cover and kept from rotating by means of lock washers.

3.3.4 Identification marking.

3.3.4.1 Stock. Legibly engraved on the handle of the gun stock in the area designated on sheet 1 of the drawing (coordinate C-7), there shall appear the following information:

Hypodermic Injection Apparatus, Jet, Automatic
Serial Number _____
Manufacturer's name or registered trademark

The serial numbers shall run consecutively for each contractor, no numbers being repeated. The lettering shall be approximately 3/32 inch high and the engraving shall be filled with a white lacquer filling after the stock is anodized. In addition, the stock shall have engraved on each side, near the observation windows, the dosage calibration marks as shown on sheet 6 of the drawing.

3.3.4.2 Carrying case. The carrying case shall be fitted with two anodized aluminum-alloy nameplates having a thickness of 0.020 inch. One nameplate (part 19) shall measure approximately 2 by 5/8 inches and shall bear the following legend: "REPAIR PARTS". This nameplate shall be located on the outer repair parts compartment cover, secured by two rivets which simultaneously secure two of the spare injection mounts (part 47) in place. Another nameplate of suitable size to fit within the recessed area of the carrying case top shall bear the following information:

"Hypodermic Injection Apparatus, Jet, Foot Powered
NSN 6515-00-910-0097
Manufacturer's name or registered trademark
Model No. _____
Unit Serial No. _____
The medical services symbol, as shown for type III,
Standard MS 16686"

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Other information, such as applicable patent numbers, etc., may be added at the manufacturer's discretion. Letters on all nameplates shall be Gothic capitals of appropriate size. Characters shall be integrated into the plate by a photographic process. Plates shall be of a reverse type, having a black background. After the photographic processing, the image shall be sealed in the oxide layer by a suitable chemical treatment of the layer. The second nameplate shall be fastened to the case by means of Pliobond adhesive. This nameplate shall be secured to the outside of the case, centered in the case recess. The case surface shall be properly prepared to insure adherence of the nameplate.

3.3.5 Finish.

3.3.5.1 Machine finish symbols. Machine finish symbols shall be as defined in ANSI Standard B-6.1. Machine stainless steel parts shall be cleaned ultrasonically, and passivated, as specified in 3.12. Ultrasonic cleaning shall not leave a detergent film.

3.3.5.2 Black anodizing. Aluminum-alloy components which are specified to have a black anodized finish shall be finished in a heavy, uniform continuous black anodic film, free of breaks, scratches, and other damage. The gun stock (part 11) shall be given a grained satin finish (approximately 280 grit), prior to anodizing. Immediately after anodizing, and before handling, anodized parts shall be waxed to protect the finish. Anodic coating shall comply with MIL-A-8625 type II, black-dyed coating.

3.3.5.3 Vinyl plastisol coating. Specified surfaces of the bottle holder assembly shall be coated with a black or gray vinyl plastisol coating. The coating shall be accomplished through the use of Unit Chromium Division, Metal & Thermit Corporation, Unichrome Coating 5300 (Black or Grey) applied on a coat of Unichrome Primer 219-PX. The coating shall be applied following the manufacturer's recommended coating procedure to yield a final backed film having a minimum thickness of 5-10 mils. Finished coat shall be a tough, resilient coating free of pin-holes and blisters.

3.3.5.4 Trigger. The cast trigger (part 20) shall be polished to round all edges and corners and then given a heavy chromium plating over nickel plating.

3.3.5.5 Stainless steel parts. All exterior exposed stainless steel parts shall be given a No. 4 or better uniform satin finish. This includes part numbers 7, 8, 26, 28, 29, 30, 33, 34, 35 and 41.

3.3.5.6 Hardware, fasteners, etc. Except when otherwise specified, all hardware, fasteners, hinges, etc., that are used as components of the plastic carrying case and that are of steel or brass construction shall be given a heavy nickel plating either luster or satin finish.

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3.3.6 Repair parts. With each unit there shall be furnished the repair parts listed on sheets 14 and 15 of the bill of materials accompanying the drawing. All repair parts, with the exception of the wrench, shall be placed within the repair parts compartment. The spare jet nozzle, with protective cap, shall be screwed onto one of the mounts (part 47) on the inside of the case lid. Prior to being placed into the repair parts compartment, the repair parts shall be segregated into groups, as follows

Bag No.	Contents
1	Spare feed needles and adapters, sharp end of feed needles shall be protected.
2	Springs (springs shall be mounted on a corrosion-resisting steel wire loop)
3	Balls
4	Seals and seal back-up ring
5	Nozzle cleaning adapters (see 3.3.1.6.1.2)

Each group shall be neat sealed in a clear plastic envelope. All sealed envelopes shall then be loaded into the repair parts compartment with the high mortality parts list. This parts list shall itemize all components by bag number, part number quantity and description.

3.3.7 Weight. The finished weight of the jet injector within the carrying case, including repair parts and manuals, shall not exceed 12-1/2 pounds.

3.3.8 Assembly. The hydraulic power unit on the cast mount shall be slipped under the angle retainer, part 5, and secured in place by the rear screw entering the eyeletted opening of the carrying case. The hoses shall be wrapped counterclockwise around the repair parts compartment of the case, using the cover as a hose retainer. The injection gun shall be clipped into the case cover, with the identification marking shown. The securing strap assembly (part 9) shall be buckled shut. The final assembly shall be as pictured on sheet 1 of the drawing and in a photograph included in the operating instruction manual (see 3.7). Hoses shall be oriented following a procedure given in the manual so that they can be wrapped into the case without kinks or twists.

3.4 Government-furnished material. No Government-furnished material will be provided, however, the following items will be made available by the Government to the Government inspector for use in inspecting subject item.

(a) One set of gages for checking the most critical dimensions of the gun stock (part 11).

(b) One set of adapters for: (1) the inlet valve test (4.3.2.1); (2) the outlet valve test (4.3.2.2); and (3) the feed needle test (4.3.3).

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At the conclusion of the contract, the contractor shall carefully check the gages and adapters, recondition them as necessary, and repack them for transport in their reusable carrying cases. Instructions for the use of the gages are included in the carrying cases.

3.5 Tooling. The contractor shall design and manufacture or provide any tooling needed for the production and testing of hypodermic injection apparatus as required to meet specification requirements described herein. Such tooling shall remain the property of the contractor.

3.6 The contractor shall perform all services, as specified, and will not produce any of said hypodermic injection apparatus other than in strict accordance with requirements contained in Section 3 of this specification without the previous written approval of the contracting officer.

3.7 Technical manual. The contractor shall prepare and furnish 15 technical manuals, plus two for each equipment set (packed in a manila envelope placed in the carrying case). The technical manuals shall be type I, in accordance with MIL-M-15071, except approval shall be by Defense Personnel Support Center. It should be noted that manuals shall be completed in time for Defense Personnel Support Center review and approval, and are to accompany each shipment. Manuals not accompanying equipment shall be forwarded to the contracting officer prior to completion of the contract. A list of all component repair parts subject to wear and replacement, identified by their manufacturer's name, address, model or catalog number, and current prices shall be included. An apron sheet (fold-out) of the jet gun assembly shall be provided at the rear of the manual for ready reference during review of the manual. The apron sheet shall be Kimberly Clark's No. 115, Textopoint paper, or equal.

3.8 Supplier responsibility for maintenance. The supplier shall, for a period of at least two years after completion of the contract, supply under contract or purchase order, such repair parts as are peculiar to his manufacture, or other commercial parts modified by him to provide components for the hypodermic injection apparatus supplied by him.

3.9 Bill of materials. There shall be provided a final Government bill of material covering the hypodermic injection device as required by the schedule.

3.10 Interchangeability. The hypodermic injection apparatus shall be manufactured in such a manner that all removable component parts shall be interchangeable with corresponding parts in any apparatus produced under this specification.

3.11 Tests. The hypodermic injection apparatus shall perform in accordance with the following requirements.

3.11.1 Clinical tests. If deemed desirable (if contractor has not previously supplied acceptable injectors), the Government will arrange for the clinical testing of the hypodermic injection apparatus on patients using one of the preproduction samples in order to insure the proper functioning of the sample, including injecting with a minimum of whealing or bleeding and other characteristics described herein.

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3.11.2 Leakage test, hydraulic gun. The hydraulic system shall pass the leakage test described in 4.3.1.

3.11.3 Leakage test, air. The vaccine pump section shall meet the vaccine pump section test described in 4.3.2.

3.11.4 Leakage test, feed needle. The feed needle assembly shall meet the air leakage test described in 4.3.3.

3.11.5 Jet nozzle test. The sapphire jewel orifice shall comply with the dimensional requirements of the drawing, and, on firing, there shall be no leakage or side spray per test described in 4.3.4.

3.11.6 Hydraulic power unit test. Hydraulic power units shall comply with the tests described in 4.3.5.

3.11.7 Performance test, hypodermic injection apparatus. The injection apparatus shall be test fired as described in 4.3.6. The injection apparatus shall be capable of sustained continuous performance as described in 4.3.6 without failure of any sort.

3.11.8 Vibration test. The injection gun shall not be shaken loose from the supporting clips (part 7 and 8) or damage occur to the securing strap (part 9) when unit is subjected to vibration test specified in 4.3.7. This test may be waived upon request by contractor, when contractor has previously furnished items in accordance with requirements of this specification.

3.12 Cleaning. Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice. Cleaning processes shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After machining, all stainless steel parts including the assembled jet nozzle shall be ultrasonically cleaned and passivated. After assembly, components shall be cleaned throughly and shall be free from particles of filler metal, flux, and other foreign material. If necessary, such cleaning shall also be performed after final assembly of components. Prior to packaging, the plastic carrying case shall be rubbed down with a wax-laden cloth.

3.13 Workmanship. Workmanship shall be first class throughout. Hypodermic injection apparatus shall be free of defects which detract from their appearance or may impair their serviceability.

3.13.1 Joining. Brazing shall be in accordance with MIL-B-7883. Brazed joints shall be durable and sound in every detail. Component parts shall be carefully and properly aligned prior to brazing into permanent position. Brazed joints shall be polished smooth to blend with adjacent surfaces Spot welds shall be free of surface burning, cracks, and porosity, and shall be tightly fused.

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3.13.2 Machine work. All component parts shall be free of burrs and other imperfections. All threads on threaded members shall be properly formed and of the fit specified in the drawing. Ball seats shall be properly formed and burnished to assure leaktight sealing of the balls. Alignment of components shall be such that assembly and disassembly shall be easily accomplished and such that the function of the injector is not impaired, due to uneven wearing or scratching of components caused by misalignment.

3.13.3 Hoses. Hydraulic hoses shall be clear of any rubber or foreign particles before being assembled in the injection apparatus by purging with oil (see 3.3.2.1.5). Woven covering shall be free of fraying. Hoses shall be attached so that they may be neatly coiled when the injection gun is mounted within the carrying case.

4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, of any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means or physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the components or materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the component and material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements

4.2 Examination. The hypodermic injection apparatus shall be examined to determine that the materials, design, construction, physical dimensions, and workmanship are in accordance with applicable requirements. Visual and mechanical defects shall be classified during the course of the contract.

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4.3 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.3.1 Leakage test, hydraulic valves, and seals. A completely assembled injector unit except for vaccine pump section, feed needle and bottle holder, shall be adjusted so that the dosage setting is at the 1.0 cc mark when the spring is fully cocked. The unit shall be left in the cocked position for a period of 15 minutes. At the end of this period of time, there shall be no discernible creeping of the hydraulic piston as seen through the observation window. If creeping occurs, a faulty firing valve, check valve or quad seal is indicated. Manufacturer will be permitted to repair unit and resubmit it for retest.

4.3.2 Vaccine pump section test. The vaccine pump action shall be subjected to the following air pressure tests to determine leakage of valves, seals, or brazed joints.

4.3.2.1 Inlet valve feed tube, brazed joints, seals (parts 93, 94, 96) test. An adapter, of aluminum alloy or other suitable metal, shall be fabricated by the contractor to permit air pressure to be applied to the inlet end of the vaccine pump cylinder (part 33), utilizing the coupling nut (part 34) to attach this adapter to the cylinder. A metal shim shall be placed in the injection nozzle blank (part 41) so that no air can leak out of the sapphire jewel orifice. If desired, a nozzle blank with no opening for the orifice may be used in this test. The vaccine pump assembly shall be filled with water, attached to air supply and then held vertically, nozzle down, immersed up to the coupling nut in water. An air pressure of 80 psi shall be applied to the assembly. There shall be no evidence of air leakage past the injection nozzle, feed tube joints, or inlet valve. A defective vaccine pump assembly may be repaired and submitted for retest.

4.3.2.2 Outlet valve leak test. An adapter of aluminum alloy, or other suitable metal, shall be fabricated by the contractor to permit air pressure to be applied in reverse, or nonflow direction, to the outlet valve. This adapter shall be similar to the jet nozzle mount, except that the orifice shall be replaced by an air inlet fitting. The adapter shall screw onto the threads of the vaccine cylinder (part 33) and shall employ the teflon seal for sealing purposes. The outlet valve shall be immersed in water and an air pressure of 15 psi shall be applied to the outlet valve. There shall be no air leakage, as evidenced by air bubbles within the cylinder, as a result of this test. A defective outlet valve may be repaired and submitted for retest.

4.3.3 Feed needle test.

4.3.3.1 Leakage - air. An adapter, of aluminum alloy or other suitable metal, shall be fabricated by the contractor to permit air pressure to be applied to the inside of the hub (part 84) of the feed needle assembly. Air pressure shall be applied to check that the feed ports are open and that the feed needle annular passageway is open. Similarly, air pressure shall be applied to hub (part 83) to check that the air vent line is open. The feed needle shall be immersed in water, air pressure of 30 psi shall be applied to hub (part 84), and the outlet ports shall be sealed by closing them off by means of the test operator's

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fingers. There shall be no air leakage past the brazed joints of the feed needles as a result of this test. Similarly, the brazed joint in the air vent line shall be checked by applying air pressure at hub (part 83), and sealing off the air vent needle opening.

4.3.3 Restriction test. To insure that annular space between needles is free of foreign matter (flux, braze, material, etc.), each needle shall be placed on an operating unit and used. The flow rate through the needle shall be such that the vaccine pump is not starved of solution when fired at the rate of 1200 shots per hour and the dosage adjustment is set at the 1 cc setting.

4.3.4 Jet nozzle tests. The sapphire jewel orifice contours shall be checked by means of contour projection at 100 magnifications (using diiodomethane) to determine dimensional compliance with the detail drawing. After mounting of the sapphire, the jet nozzle shall be placed on an injection device and the injector fired a minimum of 25 times. The stream emitted shall be free from all side spray. There shall be no tracer of fluid leakage around the periphery of the orifice insert or any loosening of the orifice insert in the mount. Each jet nozzle assembly, including the spare, shall be tested as described above.

4.3.5 Hydraulic acceptance test. Random samples of the hydraulic power unit or power unit components shall be tested on a manufacturer's test stand for compliance with this specification. The contractor shall make available his or his supplier's test stand capable of performing the tests specified herein for this purpose. Units which fail may be reworked and resubmitted for retest. During this test, the hydraulic pump shall be capable of generating a minimum of 3000 psi with the hydraulic fluid temperature at 100°F maximum. Similarly, the relief valve shall be capable of operating in the 500 to 2300 psi range without damage. The completely assembled unit shall be capable of operating in ambient temperatures ranging from - 30°F to + 110°F. With an injection gun connected to the power unit, the gun shall be operated a minimum of ten times to show that the pump is functioning properly and that relief valve is properly set. The unit shall be rotated about the pedal axis and shall be turned end for end without any trace of hydraulic fluid leakage. The relief valve shall be set and checked to determine that it is at its specific setting prior to assembly of the power unit to the injection apparatus.

4.3.5.1 Foot power unit tests.

4.3.5.1.1 Force requirements. The reservoir of the foot pump shall be filled with the specified hydraulic fluid and the relief valve adjusted to the pressure values given in the tabulation in paragraph 3.3.2.1.1. The maximum force at the start of the stroke and the maximum force during the stroke shall be measured and tabulated for each of the specified relief valve settings. The measured force shall not exceed those specified in paragraph 3.3.2.1.1. A total force of 250 pounds shall be applied to the pedal and the unit examined for evidence of damage.

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4.3.5.1.2 Flow requirements. The relief valve of the power unit shall be set at 1300 psi ± 50 psi. A second relief valve set at 1100 psi ± 50 psi shall be attached to the outlet port and the pump stroked several times to eliminate all air. The reservoir shall be filled to the designated level. There shall be no return of fluid to the reservoir. The pump shall be stroked sixteen times, and the hydraulic fluid blown through the outlet relief valve shall be collected and measured to determine compliance with delivery requirements per strokes (see 3.3.2.1.1.). The pump shall then be operated and more fluid collected to determine compliance with the amount of usable hydraulic fluid contained in the reservoir (see 3.3.2.1.2).

4.3.5.1.3 Relief valve leakage test. With the relief valve set at 1300 psi ± 50 psi, an injection gun which has met all the specification requirements shall be attached to the hydraulic power unit, the unit filled with the specified hydraulic fluid to the proper level and all air purged from the injector and hoses. The injection gun dosage control shall be set at 0.5 cc. The foot powered pump shall then be actuated at different stroke rates and the dosage indicated at the observation window observed. All readings shall be consistant, regardless of stroke rate. If the reading at slow pumping rates is low, the relief valve is leaking and requires repairs. Relief valves may be reworked and resubmitted for test.

4.3.6 Performance test, hypodermic injection apparatus.

4.3.6.1 Leakage air test. The complete jet injection apparatus shall be set up to fire a bottle of water, following instructions given in the operator's technical manual. Water may be supplied from a continuous source. The injection apparatus shall be capable of sustained continuous performance. The injection apparatus shall be mounted in a test jig designed to fire the apparatus at the rate of 1200 shots per hour. Each unit manufactured shall be fired on a test stand for at least 750 shots. The unit under test shall fire at least 250 1 cc shots and 500 ½ cc shots of water continuously without failure at the rate of 1200 shots per hour. In addition, one gun from each production lot shall be fired continuously for 5000 shots with the dosage set at 1 cc. There shall be no evidence of leakage or malfunctioning at the conclusion of the test. With dosage set at 1.00 cc, the vaccine pump shall purge itself of air and operate smoothly after 3 to 4 "shots". The vaccine stream shall be straight, without side streams, and shall cut off sharply at the end of the shot. There shall be no trailing which indicates the presence of air in the vaccine pump. All valves shall operate smoothly, and there shall be no hydraulic leakage past the gun stock valves, the observation window, the rear dosage adjustment knob, and the hydraulic hose connections to the gun. On cocking the gun, there shall be no drawback of fluid at the jet nozzle. After the gun is cocked, the solution bottle shall be removed, and the gun fired. There shall be no more than one drop of liquid forced back through the vaccine pump inlet valve. The dosage adjustment device shall be checked for ease of adjustment and the unit shall be operated through several cycles at 0.2 and 0.5 cc doses without failure. Note: One complete turn of the adjustment knob is equivalent to a 0.1 cc change in dose.

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4.3.6.2 Feed needle assembly test. In addition to the air test, each feed needle assembly (including spares) shall be subjected to a minimum of 25 shots without sign of any leakage, and with proper and sufficient flow of solution. The brazed points at either end of the inlet tube assembly (part 35) shall show no leakage at the conclusion of the performance test. Power for driving the gun can be supplied by the hydraulic pump itself or by the test stand.

4.3.7 Vibration test. The jet injection apparatus shall be installed in its carrying case and the securing strap (part 9) shall be pulled tightly to secure the gun into the gun clips. The case shall be closed and the unit placed unconfined in a horizontal position, on the four case bumpers, on the platform of vibration package tester, LAB Corporation or equal, which has a total throw or displacement of 1 inch. The tester shall be operated in non-synchronous motion, i.e. circular motion combined with table tipping. The platform shall be operated for a minimum of 5 minutes at a frequency of 5 cycles per second. At no time during this vibration procedures should the injector leave the confines of the gun clips, or damage occur to the case or the securing strap assembly.

4.4 Government inspection. The contractor warrants that the contracting officer or his authorized representative shall, for the purpose of inspection, have access at all reasonable times to any plant(including the plant of any subcontractor)in which any of the work is being performed. In addition to inspection and testing required of the contractor, it is intended that in-process and final inspection will be conducted by the Naval Inspection Service under the technical direction of the Defense Personnel Support Center, Directorate of Medical Materiel. First article inspection will be conducted by the Government.

4.4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Prior to the submission of each lot of the injection apparatus for acceptance by the Government, the contractor will conduct and document a thorough physical examination of each unit, and make and record operational tests described herein to demonstrate conclusively the proper functioning of each unit. This inspection shall include all examinations and tests described herein as well as any other tests found necessary during the production of the unit, including but not limited to the following criteria:

(a) Acceptance of Preproduction Sample.
(b) Approval of Materials and Components.
(c) Visual and Mechanical Examination.
(d) Compliance with Drawing and Bill of Materials.
(e) Approval of Clinical Test, (if performed).
(f) Performance Test.
(g) Other Tests described in 3.11.

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Subsequent to the contractor's inspection as indicated above, an examination of the units by the authorized Government representative shall be made. Upon failure of any unit ot perform, or of any component, the contractor shall replace the faulty unit or faulty component with new material. Upon completion of the testing by the Government representative as indicated above, the contractor shall carefully examine all units to insure proper condition, cleanliness, and preparation for shipment. Packaging and packing snall be included as part of this inspection.

4.5 Sampling. Sampling for material or components snall be conducted by the Government in accordance with MIL-STD-105, as appropriate. Testing and examination of finished injection apparatus shall be on a 100% basis, except tnat the endurance portion of the performance test (see 3.11.7 for 5000 shot run) shall apply to one of each production lot, or to one of each fifty unit, wnicnever is smaller.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. Each hypodermic apparatus, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Components snall be positioned and secured witnin the carrying case in such a manner as to prevent shifting or damage. The carrying case snall be furtner packaged in accordance with MIL-P-116, method IA-15 or IC-2 except tnat a preservative shall not be utilized.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, wnicn will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Each unit shall be overboxed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant, grade V3c or V3s, style RSC. Closure, strapping, and waterproofing snall be as specified in PPP-B-636

5.2.2 Level B.

5.2.2.1 Exterior container. Each unit shall be overboxed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic and fabricated of not less than 200 lb. test fiberboard. Method II closure as specified in PPP-B-636 shall apply.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following

(a) Number and date of this specification.

(b) Selection of applicable levels of packing and packaging (see 5.1 and 5.2).

(c) National Stock Number.

6.2 This specification covers NSN 6515-00-910-0097 HYPODERMIC INJECTION APPARATUS, JET, AUTOMATIC, Foot Operated listed in the Federal Supply Catalog.

6.3 This specification does not include all types, classes, sizes or grades as indicated by the title of this specification nor all those which are commercially available, but is only intended to cover the types which are normally procured to meet Military requirements.

Custodians
Army - MD
Navy - MS
Air Force - 03

Preparing activity
Defense Supply Agency - DM

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DOCUMENT IDENTIFIER AND TITLE

MIL-E-37084 HYPODERMIC INJECTION APPARATUS, JET, FOOT OPERATED

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE

☐ YES ☐ NO *(If "Yes" in what way?)*

4 REMARKS

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